Question 1

Not yet answered

Marked out of 1.00

Given:

```
3. public class Ebb {
4. static int x = 7;
5. public static void main(String[] args) {
6. String s = "";
7. for(int y = 0; y < 3; y++) {
8. x++;
9. switch(x) {
10. case 8: s += "8 ";
11. case 9: s += "9 ";
12. case 10: { s+= "10 "; break; }
13. default: s += "d ";
14. case 13: s+= "13 ";
15. }
16. }
17. System.out.println(s);
18. }
19. static { x++; }
20. }
```

What is the result?

Select one:

- **a. 8 9 10 9 10 10 d 13**
- **b. Compilation fails**
- **c. 9 10 10 d 13**
- **d. 9 10 10 d**
- **e. 8 9 10 10 d 13**
- **f. 9 10 d**
- **g. 8 9 10 d**

Question 2

Not yet answered

Marked out of 1.00

Predict the output of the below C program:

```
#include <stdio.h>
#define SIZE(arr) sizeof(arr) / sizeof(*arr);
void fun(int* arr, int n)
{
int i;
*arr += *(arr + n - 1) += 10;
}

void printArr(int* arr, int n)
{
int i;
for(i = 0; i < n; ++i)
printf("%d ", arr[i]);
}

int main()
{
int arr[] = {10, 20, 30};
int size = SIZE(arr);
fun(arr, size);
printArr(arr, size);
return 0;
}
```

Select one:

- **a. 50 20 40**
- **b. 20 20 40**
- **c. 20 30 40**
- **d. Compile-time error**

Question 3

Not yet answered

Marked out of 1.00

Predict output of following C program

```
int main()
{
int i;
int arr[5] = {1};
for (i = 0; i < 5; i++)
printf("%d ", arr[i]);
return 0;
}
```

Select one:

- **a. 1 0 0 0 0**
- **b. 0 0 0 0 0**
- **c. 1 followed by four garbage values**
- **d. 1 1 1 1 1**

Question **4**

Not yet answered

Marked out of 1.00

Consider the following declaration of a ‘two-dimensional array in C:

char a[100][100];

Assuming that the main memory is byte-addressable and that the array is stored starting from memory address 0,
the address of a[40][50] is

Select one:

- **a. 5050**
- **b. 4050**
- **c. 5040**
- **d. 4040**

Question **5**

Not yet answered

Marked out of 1.00

A relational schema for a train reservation database is given below.
Passenger (pid, pname, age)
Reservation (pid, class, tid)

Table: Passenger
pid pname age
---
0 Sachin 65
1 Rahul 66
2 Sourav 67
3 Anil 69

Table : Reservation
pid class tid
---
0 AC 8200
1 AC 8201
2 SC 8201
5 AC 8203
1 SC 8204
3 AC 8202

What pids are returned by the following SQL query for the above instance of the tables?

```
SLECT pid
FROM Reservation ,
WHERE class ‘AC’ AND
EXISTS (SELECT *
FROM Passenger
WHERE age > 65 AND
Passenger. pid = Reservation.pid)
```

Select one:

- **a. 1,3**
- **b. 1,2**
- **c. 1,5**
- **d. 1,0**

Question 6

Not yet answered

Marked out of 1.00

Consider the function f defined below.

```
struct item
{
int data;
struct item * next;
};

int f(struct item *p)
{
return (
(p == NULL) ||
(p->next == NULL) ||
(( P->data <= p->next->data) && f(p->next))
);
}
```

For a given linked list p, the function f returns 1 if and only if

Select one:

- a. **not all elements in the list have the same data value.**
- b. **the elements in the list are sorted in non-decreasing order of data value**
- c. **None of them**
- d. **the elements in the list are sorted in non-increasing order of data value**

Question 7

Not yet answered

Marked out of 1.00

In C, fseek() should be preferred over rewind() mainly because

Select one:

- a. **rewind() doesn't work for empty files**
- b. **In rewind, there is no way to check if the operations completed successfully**
- c. **All of the above**
- d. **rewind() may fail for large files**

Question 8

Not yet answered

Marked out of 1.00

Given:
3. class Infinity { }
4. public class Beyond extends Infinity {
5. static Integer i;
6. public static void main(String[] args) {
7. int sw = (int)(Math.random() * 3);
8. switch(sw) {
9. case 0: { for(int x = 10; x > 5; x++)
10. if(x > 10000000) x = 10;
11. break; }
12. case 1: { int y = 7 * i; break; }
13. case 2: { Infinity inf = new Beyond();
14. Beyond b = (Beyond)inf; }
15. }
16. }
17. }
And given that line 7 will assign the value 0, 1, or 2 to sw, which are true? (Choose all that apply.)

Select one or more:

- ☐ **a. A StackOverflowError might be thrown**
- ☐ **b. An IllegalStateException might be thrown**
- ☐ **c. The program will always complete without exception**
- ☐ **d. The program might hang without ever completing**
- ☐ **e. A ClassCastException might be thrown**
- ☐ **f. Compilation fails**
- ☐ **g. A NullPointerException might be thrown**

Question 9

Not yet answered

Marked out of 1.00

What is Granularity?

Select one:

- ○ **a. The size of record**
- ○ **b. The size of data items**
- ○ **c. The size of database**
- ○ **d. The size of file**

Question **10**

Not yet answered

Marked out of 1.00

Consider the following relational schema:

Suppliers(sid:integer, sname:string, city:string, street:string)
Parts(pid:integer, pname:string, color:string)
Catalog(sid:integer, pid:integer, cost:real)

Consider the following relational query on the above database:

```
SELECT S.sname
FROM Suppliers S
WHERE S.sid NOT IN (SELECT C.sid
FROM Catalog C
WHERE C.pid NOT IN (SELECT P.pid
FROM Parts P
WHERE P.color<> 'blue'))
```

Assume that relations corresponding to the above schema are not empty.
Which one of the following is the correct interpretation of the above query?

Select one:

- **a. Find the names of all suppliers who have supplied a non-blue part.**
- **b. Find the names of all suppliers who have supplied only blue parts.**
- **c. Find the names of all suppliers who have not supplied a non-blue part.**
- **d. Find the names of all suppliers who have not supplied only blue parts.**

Question **11**

Not yet answered

Marked out of 1.00

Is it fine to call delete twice for a pointer?

```
#include<iostream>
using namespace std;

int main()
{
int *ptr = new int;
delete ptr;
delete ptr;
return 0;
}
```

Select one:

- **a. Yes**
- **b. No**

Question 12

Not yet answered

Marked out of 1.00

Consider the table employee(empId, name, department, salary) and the two queries Q1 ,Q2 below.

Assuming that department 5 has more than one employee, and we want to find the employees who get
higher salary than anyone in the department 5,

which one of the statements is TRUE for any arbitrary employee table?

```
Q1 : Select e.empId
From employee e
Where not exists
(Select * From employee s where s.department = "5" and
s.salary >=e.salary)
Q2 : Select e.empId
From employee e
Where e.salary > Any
(Select distinct salary From employee s Where s.department = "5")
```

Select one:

- **a. Both Q1 and Q2 produce the same answer.**
- **b. Neither Q1 nor Q2 is the correct query**
- **c. Q2 is the correct query**
- **d. Q1 is the correct query**

Question 13

Not yet answered

Marked out of 1.00

Given:

```
3. public class Circles {
4. public static void main(String[] args) {
5. int[] ia = {1,3,5,7,9};
6. for(int x : ia) {
7. for(int j = 0; j < 3; j++) {
8. if(x > 4 && x < 8) continue;
9. System.out.print(" " + x);
10. if(j == 1) break;
11. continue;
12. }
13. continue;
14. }
15. }
16. }
```

What is the result?

Select one:

- **a. 5 5 7 7**
- **b. 1 3 9**
- **c. 1 1 3 3 9 9**
- **d. 1 3 3 9 9**
- **e. 1 1 1 3 3 3 9 9 9**
- **f. Compilation fails**

Question **14**

Not yet answered

Marked out of 1.00

Consider the following C program

```
int main()
{
int x, y, m, n;
scanf ("%d %d", &x, &y);
/* x > 0 and y > 0 */
m = x; n = y;
while (m != n)
{
if(m>n)
m = m - n;
else
n = n - m;
}
printf("%d", n);
}
```

What does the program compute?

Select one:

- **a. the greatest common divisor of x and y**
- **b. x mod y using repeated subtraction**
- **c. x + y using repeated subtraction**
- **d. the least common multiple of x and y**

Question **15**

Not yet answered

Marked out of 1.00

What is time complexity of fun()?

```
int fun(int n)
{
int count = 0;
for (int i = n; i > 0; i /= 2)
for (int j = 0; j < i; j++)
count += 1;
return count;
}
```

Select one:

- **a. O(n^2)**
- **b. O(nLognLogn)**
- **c. O(n)**
- **d. O(nLogn)**

Question **16**

Not yet answered

Marked out of 1.00

Given:

```
class Emu {
static String s = "-";
public static void main(String[] args) {
try {
throw new Exception();
} catch (Exception e) {
try {
try { throw new Exception();
} catch (Exception ex) { s += "ic "; }
throw new Exception(); }
catch (Exception x) { s += "mc "; }
finally { s += "mf "; }
} finally { s += "of "; }
System.out.println(s);
} }
```

What is the result?

Select one:

- **a. -ic mf of**
- **b. -mc mf**
- **c. -ic of**
- **d. -ic mc of mf**
- **e. Compilation fails**
- **f. -mf of**
- **g. -ic mc mf of**

Question 17

Not yet answered

Marked out of 1.00

What happens when a function throws an error but doesn't specify it in the list of exceptions it can throw.

For example, what is the output of following program?

```
#include <iostream>
using namespace std;

// Ideally it should have been "int fun() (int)"
int fun()
{
throw 10;
}

int main()
{
try
{
fun();
}
catch (int )
{
cout << "Caught";
}
return 0;
}
```

Select one:

- **a. None of these**
- **b. Segmentation fault**
- **c. No compiler Error. Output is "Caught"**
- **d. Compiler Error**

Question **18**

Not yet answered

Marked out of 1.00

Given:

```
3. class SubException extends Exception { }
4. class SubSubException extends SubException { }
5.
6. public class CC { void doStuff() throws SubException { } }
7.
8. class CC2 extends CC { void doStuff() throws SubSubException { } }
9.
10. class CC3 extends CC { void doStuff() throws Exception { } }
11.
12. class CC4 extends CC { void doStuff(int x) throws Exception { } }
13.
14. class CC5 extends CC { void doStuff() { } }
```

What is the result? (Choose all that apply.)

Select one:

- ◯ **a. Compilation fails due to an error on line 10**
- ◯ **b. Compilation succeeds**
- ◯ **c. Compilation fails due to an error on line 8**
- ◯ **d. Compilation fails due to an error on line 12**
- ◯ **e. Compilation fails due to an error on line 14**

Question **19**

Not yet answered

Marked out of 1.00

Which is/are correct statements about primary key of a table?

Select one or more:

- ☐ **a. Primary keys can contain NULL values**
- ☐ **b. Primary keys cannot contain NULL values.**
- ☐ **c. A table can have only one primary key with single or multiple fields**
- ☐ **d. A table can have multiple primary keys with single or multiple fields**

Question **20**

Not yet answered

Marked out of 1.00

Given:
1. class Loopy {
2. public static void main(String[] args) {
3. int[] x = {7,6,5,4,3,2,1};
4. // insert code here
5. System.out.print(y + " ");
6. }
7. } }
Which, inserted independently at line 4, compiles? (Choose all that apply.)

Select one or more:

- ☐ **a. for(int y=0, int z=0; z<x.length; z++) { y = x[z];**
- ☐ **b. for(int y=0, z=0; z<x.length; z++) { y = x[z];**
- ☐ **c. for(int y : x) {**
- ☐ **d. int y = 0; for(y : x) {**
- ☐ **e. int y = 0; for(int z=0; z<x.length; z++) { y = x[z];**
- ☐ **f. for(x : int y) {**

Quiz Navigation

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